Yuanhao Liu

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EDUCATION

Southern University of Science and Technology (SUSTech) Shenzhen, China
*Bachelor of Engineering in **Communications Engineering*** *Expected: 06/2027*

- **Cumulative GPA:** 3.79/4.0 | **IELTS:** 7.0
- **Research Advisors:** Prof. Shuxiang Guo, Prof. Xiaoyang Li
    - **Technical Program Committee (TPC) Member**, *IEEE Wireless Communications and Networking Conference (WCNC)*, 2025; *Aerospace Information Perception and Intelligent Processing(AIPIP'26),2026*

PUBLICATIONS & MANUSCRIPTS

[1] **Yuanhao Liu**, Leheng Qi, Nan Sun, Xiaoyang Li, Changsheng You, Zidong Han, and Yi Gong. **Physics-Aware UAV Trajectory Design for Mission-Driven Fire Rescue Tasks**. Submitted to *ACC6G, 2026*.

[2] Miaozhang Shen, Zixu Wang, Yuyue Yang, Chunying Li, **Yuanhao Liu**, and Shuxiang Guo. **An Online Pose Self-Calibration Framework for Deformable Magnetmeter Arrays to Enable Accurate Capsule Endoscope Tracking**. *IEEE Transactions on Instrumentation and Measurement*, 2025.

[3] Zixu Wang, Miaozhang Shen, **Yuanhao Liu**, and Shuxiang Guo. **A Novel Flexible Soft Robotic-bandage for Gastrointestinal Wound Coverage Function**. *IEEE International Conference on Mechatronics and Automation (ICMA)*, 2025.

[4] Leheng Qi, **Yuanhao Liu**, and Xiaoyang Li. **Safety-Prioritized ISAC Beamforming for Human-Robot Collaborative Smart Factories**. Manuscript in Preparation, 2026.

RESEARCH EXPERIENCE

ICElab, SUSTech Shenzhen, China
Undergraduate Researcher, Advisor: Prof. Xiaoyang Li

Project: Physics-Aware UAV Trajectory Design for Mission-Driven Fire Rescue | First-author *11/2025-Present*

- Constructed a fire-specific wireless environment model integrating thermal-plume aerodynamics, mmWave smoke attenuation, receiver noise, and rescue-unit mobility.
- Formulated long-term Value of Information (VoI) minimization using Lyapunov optimization and developed a receding-horizon Fire-Aware Trajectory Model Predictive Control (FAT-MPC) framework.
- Designed an ATRI-SCA-based solver with penalty binary relaxation for joint 3D UAV trajectory and user-association optimization under dynamic safety constraints; in a 28-GHz multi-UAV network, achieved sub-second per-slot optimization while maintaining VoI below 10.

Project: Safety-Prioritized ISAC Beamforming for Smart Factories *07/2026-Present*

- Formulated semantic sensing QoS tiers (human > robot ≥ obstacle) for ISAC downlink beamforming, jointly considering human-safety priorities and per-robot communication-rate constraints.
- Developed semidefinite-relaxation, robust S-procedure, SCA, and CRB-based formulations to optimize transmit power and characterize sensing-communication trade-offs.
- Demonstrated that prioritized human sensing can be guaranteed with only a small transmit-power premium over uniform sensing requirements.

Guolab, SUSTech Shenzhen, China
Undergraduate Researcher, Advisor: Prof. Shuxiang Guo

Project: Physics-Informed Residual Network for Magnetic Dipole Model Correction *03/2025-12/2025*

- Built a dual-branch physics-informed residual network combining convolutional features, pose-aware physics encoding, gated fusion, and a divergence-free magnetic-field regularizer.
- Integrated magnetic-field correction with Levenberg-Marquardt pose estimation and lookup-table compensation for high-accuracy sim-to-real localization.
- Reduced mean positioning error from 2.10 to 1.15 mm (45.2%) and orientation error from 3.23° to 1.01° (68.7%), while maintaining 22.4 FPS real-time performance.

Project: Differential Magnetic Sensing with Dynamic Background Calibration *09/2025-11/2025*

- Developed differential magnetic sensing for common-mode interference suppression and Kalman-filter-based dynamic background estimation in time-varying environments.
- Applied a distance-adaptive weighted LM-GN hybrid optimizer to balance near-field model error and far-field convergence for capsule-robot localization.
- Achieved 2.52 mm RMS positioning and 2.97° orientation accuracy on a 5×5 triaxial magnetometer array at 200 Hz, reducing positioning error by 59.4% and accelerating convergence by 40.6% over conventional baselines.

Project: Kresling-Origami Soft Capsule Robot *10/2025-Present*

- Developed Kresling capsule geometry and mechanical models in SolidWorks and investigated magnetically actuated gait sequences for confined gastrointestinal environments.
- Participated in prototype fabrication and motion-characterization experiments.

SELECTED PROJECTS

Huawei Crowdsourced Offline High-Precision Localization Challenge Shenzhen, China

Algorithm Development Project *2026*

- Modeled noisy crowdsourced BLE observations using RSSI, scan-position accuracy, confidence, location source, device identity, and temporal information, generating robust location candidates through path-loss models, pairwise RSSI ranking, and geometric-median estimation.
- Developed a dual-pipeline candidate ranking and fusion framework with source-specific candidates, group-wise LambdaRank, Top-3/60 m ENU fusion, and a safety gate for adaptive model selection.
- Built a leakage-free nested 5-fold OOF evaluation pipeline, achieving 89.66% (763/851) within 30 m and CEP90 = 30.86 m, reducing CEP90 by 17.6% and approaching the 90% target within three cases.

2025 CUMCM: UAV Smoke-Screen Interference Deployment Optimization Shenzhen, China

Team Member | ***National First Prize*** *09/2025*

- Developed kinematic and geometric occlusion models for UAVs, smoke-screen payloads, incoming targets, and protected objects through time-discretized physical simulation.
- Designed simulated annealing, particle swarm optimization, and hybrid PSO-SA algorithms for single- and multi-UAV deployment, increasing effective occlusion duration from 1.39 s to 20.80 s in the multi-UAV/multi-target scenario.

HONORS & AWARDS

- **National First Prize,** China Undergraduate Mathematical Contest in Modeling (CUMCM), 2025
- **Honorable Mention,** Mathematical Contest in Modeling/Interdisciplinary Contest in Modeling (MCM/ICM), 2026
- **Project Lead,** University Student Innovation and Entrepreneurship Training Program, 2026
- **Project Member,** SUSTech “Climbing Plan” Special Research Funding, 2026

SKILLS

- **Programming & Systems:** Python, MATLAB, C/C++, Linux, LaTeX, Keil
- **Algorithms:** Reinforcement Learning, Convex Optimization, Array & Sparse Signal Processing, Physics-Informed Learning
- **Engineering Tools:** SolidWorks, COMSOL, HFSS, STM32/ESP32, Raspberry Pi, OpenMV